

Assignment-2 Solution

$$1) xy'z' + x'y'z + xyz + x'yz'$$

$$= xy(z' + z) + x'y'z + x'yz'$$

$$= xy + x'y'z + x'yz'$$

$$= y(x + x'z') + x'yz'$$

$$= y(x + x')(x + z) + x'yz'$$

$$= y(x + z) + x'yz'$$

$$= xy + yz + x'yz'$$

$$= xy + y(z + x'z')$$

$$= xy + y(z + x')(z + z')$$

$$= xy + y(z' + z)$$

$$= xy + x'y + yz$$

$$= y(x + x') + yz$$

$$= y + yz \text{ (Absorption Law)}$$

$$=$$

$$= y$$

$$2) (a'b + cd)e' + e$$

$$= [(a' + b)(c + d) + e'] \cdot e$$

$$= [(a + b')(c' + d') + e] \cdot e'$$

POS

$$\begin{aligned}
 F(v, w, x, y, z) &= wy + wx + x'y \\
 &= w(x+y) + x'y \\
 &= (w+x'y)(x+y+x'y) \\
 &= (w+x')(w+y)(x+y+x') \\
 &= (w+x'+v.v')(w+y+v.v')(x+y+v.v') \\
 &= (v+w+x')(v'+w+y)(v'+w+y)(v+x+y)(v'+x+y) \\
 &= (v+w+x'+y.y')(v'+w+x'+y.y')(v+w+y+x.x')(v'+w+y+x.x') \\
 &\quad (v+x+y+w.w')(v'+x+y+w.w') \\
 &= (v+w+x'+y)(v+w+x'+y')(v'+w+x'+y)(v'+w+x'+y')(v+w+x+y) \\
 &\quad (\cancel{v+w+x'+y})(v'+w+x+y)(\cancel{v'+w+x'+y})(\cancel{v+w+x+y})(v+w'+x+y) \\
 &\quad (\cancel{v'+w+x+y})(v'+w'+x+y) \\
 &= (v+w+x'+y+z.z')(v+w+x'+y'+z.z')(v'+w+x'+y+z.z')(v'+w+x'+y'+z.z') \\
 &\quad (v+w+x+y+z.z')(v'+w+x+y+z.z')(v+w'+x+y+z.z')(v'+w'+x+y+z.z') \\
 &= (v+w+x'+y+z)(v+w+x'+y+z')(v+w+x'+y'+z)(v+w+x'+y'+z') \\
 &\quad (v'+w+x'+y+z)(v'+w+x'+y+z')(v'+w+x'+y'+z)(v'+w+x'+y'+z') \\
 &\quad (v+w+x+y+z)(v+w+x+y+z')(v'+w+x+y+z)(v'+w+x+y+z') \\
 &\quad (v+w'+x+y+z)(v+w'+x+y+z')(v'+w'+x+y+z)(v'+w'+x+y+z') \\
 &= (00100, 00101, 00110, 00111, 10100, 10101, 10110, 10111, \\
 &\quad 00000, 00001, 10000, 10001, 01000, 01001, 11000, 11001) \\
 &= \pi(4, 5, 6, 7, 20, 21, 22, 23, 0, 1, 16, 17, 8, 9, 24, 25)
 \end{aligned}$$

4) $F(A, B, C, D) = (ABCD + A'D^2 + (B'+D)')^2$ using NAND

